



Review

The educational use of social annotation tools in higher education: A literature review

Elena Novak ^{a,*}, Rim Razzouk ^{a,1}, Tristan E. Johnson ^{b,2}^a Educational Psychology and Learning Systems, Florida State University, University Center C4600, Tallahassee, FL 32306, United States^b Human Performance Research Center at the Learning Systems Institute, Educational Psychology and Learning Systems, Florida State University, University Center C4600, Tallahassee, FL 32306, United States

ARTICLE INFO

Available online 29 September 2011

Keywords:

Social annotation technology
Online learning
Higher education
Literature review

ABSTRACT

This paper presents a literature review of empirical research related to the use and effect of online social annotation (SA) tools in higher education settings. SA technology is an emerging educational technology that has not yet been extensively used and examined in education. As such, the research focusing on this technology is still very limited. The literature review has aimed at presenting a comprehensive list of SA empirical studies not limited to a particular research method or study domain. Out of more than 90 articles that were initially found, only 16 studies met the inclusion criteria. Among the included studies were eight experimental or quasi-experimental studies and eight evaluation/survey studies. The SA empirical research has provided some evidence regarding the potential effectiveness of integrating social annotation tools into learning activities. Findings from the gathered literature were synthesized to provide recommendations for using SA technology in educational settings.

© 2011 Elsevier Inc. All rights reserved.

Contents

1. Introduction	39
2. Method	41
2.1. Procedure	41
2.2. Inclusion criteria	41
3. Findings	41
3.1. Evaluation/survey studies	41
3.1.1. Use of annotations for information search tasks and reading comprehension	41
3.1.2. Peer critique SA activities	43
3.1.3. Critical analysis and comprehension	44
3.1.4. Online SA-supported collaborative learning	44
3.2. Experimental and quasi-experimental studies	45
3.2.1. Reading comprehension, critical thinking, and meta-cognitive skills	45
3.2.2. Reading comprehension and emotions and motivation toward reading task	46
3.2.3. Reading comprehension	46
3.2.4. Learning achievement, attitudes, and relationships between quality of annotations and learners' characteristics	47
3.2.5. Information search	47
4. Discussion	47
5. Recommendations for using social annotation technology	48
6. Future research	48
References	49

* Corresponding author. Tel.: +1 850 570 4044.

E-mail addresses: enovak@fsu.edu (E. Novak), rr05e@fsu.edu (R. Razzouk), tejohnson@fsu.edu (T.E. Johnson).

¹ Tel.: +1 850 567 0967.² Tel.: +1 850 644 8770.

1. Introduction

Educational technology plays an important role in developing and supporting instructional activities that promote learning and facilitate students' engagement in a learning process. Social annotation

(SA) tools are an emerging educational technology that has been drawing more of educators' attention over the last several years. SA technology is an online social bookmarking tool that allows annotating (adding comments, highlights, sticky notes, etc.) of an electronic resource and supports easy online information sharing. SA technology offers knowledge sharing solutions and a social platform for interactions and discussions.

SA tools provide a means for users to modify or attach any type of content to any electronic resource (usually a text) where users can interact. Social annotation tools enable users to collaboratively underline, highlight, and annotate an electronic text, in addition to providing a mechanism to write additional comments on the margins of the electronic document (Huang, Huang, & Hsieh, 2008; Kawase, Herder, & Nejdil, 2009). The embedded annotations such as tags, comments, and highlights can be shared with other users at a time later on. A SA technology is a multi-purpose system that facilitates adding, editing, and modifying information in an electronic resource without changing the resource itself. It also allows users to share their annotations and web resources with friends or group/project members and discuss the content of a certain resource, which fosters new level of knowledge through aggregating information from many users. The annotations can be thought of as an addition layer of information on top of the existing resource that can be shared with other users using the same SA system.

SA tools are sometimes confused with Text Annotation tools such as MS Word, Adobe PDF, Google Docs. Similar to SA tools, it is possible to make comments and highlight electronic text in MS Word or Adobe PDF, for example. However, Text Annotation tools are not considered SA technology, since they do not provide an online social platform for information sharing. A user has to share manually an MS Word or Adobe PDF annotated document via email or other communication methods. Other tools provide an online social platform for information sharing but lack some of the SA technology characteristics. For example, Google Docs provides an online social support platform but does not allow annotating of newly uploaded electronic materials/documents created via other tools.

In order to facilitate discussion of educational uses of the SA technology, the following SA features have been identified. First, SA technology allows users to make written annotations such as notes or comments. Annotating is implemented by creating and attaching comments to specific sentences or areas within an electronic resource. Second, within a SA system a user can highlight any part of electronic text using either different colors or various types of notations and marks including underlining, circling, or boxing selected passages within the text. Third, SA technology provides an online platform for social collaboration. It enables creating private groups where students can collect electronic resources, annotate and highlight these resources, and share all this information with others. Moreover, students can collaboratively discuss electronic content on the margins of a digital paper and tie it back to a specific text section (s). This information sharing could be done either synchronously (i.e., instant sharing of information with others who are present/available online) or asynchronously (i.e., sharing of information while others are not available online). When information sharing is done asynchronously, the SA technology notifies about the changes made. Some SA tools enable public notes sharing and text formatting. However, the majority of the SA tools do not support the text-formatting feature yet.

The first three SA features (i.e., annotations, highlights, and collaboration/information sharing online platform) described above, define the concept of SA educational technology, and are common among all SA tools. What distinguishes one SA tool from another is its design and technical features. Glover, Xu, and Hardaker (2007) provide a list of literature-supported requirements and desirable conceptual and technical factors for SA educational systems (see Table 1). These requirements have been drawn based on the examination of the literature related to web annotation and assessment of

Table 1

Summary of essential and desirable factors for SA systems. Adapted from "Online annotation — Research and practices," by I. Glover, Z. Xu, and G. Hardaker, 2007, *Computers & Education*, 49, 1308–1320.

	Essential factors	Desirable factors
Conceptual	• Ability to add text annotations • Private annotations	• Ability to add graphical annotations (images, concept maps, etc.) • Ability to link annotations • Ability to share annotations amongst students • Ability to collate annotations into structured reports
Technical	• No additional software necessary • Compatibility with assistive technologies • Browser independent • Maintains page integrity	• Open architecture • Non-commercial

publicly available systems. According to Glover et al. (2007), the list is neither exhaustive nor absolute since differences exist among various SA tools. Nevertheless, the listed requirements provide a generic overview of the considerations that might be encountered when selecting a SA tool for education purposes.

Many annotation online tools have been developed over the last decade, including Gibeo (Bateman, Farzan, Brusilovsky, & McCalla, 2006), Annotea (Wu-Yuin, Chin-Yu, & Mike, 2007), EDUCOSM (Nokelainen, Kurhila, Miettinen, Floreen, & Tirri, 2003), Diigo (Kawase et al., 2009), HyLighter (Lebow & Lick, 2004), The Fluid Annotations projects (Zellweger, Mangen, & Newman, 2002), and MADCOW (Bottoni et al., 2004, 2006).

The growing number of social annotation technologies can be explained by the need for using such tools in various settings and the benefits this technology offers. Collaborative annotating allows collecting and organizing of resources, and accessing and sharing them with others easily from anywhere. The collaborative aspect of these tools creates a sense of community among the users in a given system and allows them to become more involved in the community (Bateman et al., 2006). The SA tools are particularly useful for teams working on projects.

SA tools have been successfully used in educational settings. Researchers claim that annotation technologies increase participation and engagement (Lebow & Lick, 2004; LeeTiernan & Grudin, 2001, as cited in Wu-Yuin et al., 2007), improve instruction (Lebow & Lick, 2004), promote attention, communication, and organization (Yang et al., 2004, as cited in Huang, et al., 2008; Davis & Huttenlocher, 1995, as cited in Wu-Yuin et al., 2007). Annotations can improve reading comprehension and peer-critique skills (Archibald, 2010; Johnson, Archibald, & Tenenbaum, 2010; Mendenhall & Author, 2010). In addition to benefiting the annotator, other readers may also benefit from reviewing shared annotated documents (Kawase et al., 2009). Readers may benefit by gaining ideas, seeing others' different perspectives, and building knowledge about the annotated resource.

However, most of the mentioned benefits have not been supported by a rigorous empirical research. There are many challenges inherent to the use of SA technologies in educational settings (Johnson et al., 2010). It is unclear how to use these tools to improve academic achievement. Do students really benefit from using SA tools? No comprehensive literature review investigating the SA use in educational settings and SA effect on academic achievement has been found.

The purpose of the paper is to review existing literature related to the (1) uses of SA in educational settings and (2) effects of SA technology on learning. A literature review of empirical studies concerning the use and the effect of social annotation tools in higher educational settings is presented in this paper. The questions underlying this review

are: (a) Does SA technology lead to any learning gains in higher education, and under what conditions does SA promote learning? (b) What are the general learning benefits and learning outcomes linked to SA? (c) How has SA technology been embedded into instructional activities and what types of learning have been facilitated the most through using SA technology?

2. Method

2.1. Procedure

A large body of literature focusing on online social annotation tools was gathered and reviewed. The following online databases were employed for the literature search: ERIC, PsychInfo, and ScienceDirect. In addition, *Google Scholar* was used to search for and acquire specific references.

2.2. Inclusion criteria

The focus of the search was to gather full-text articles presenting empirical studies employing social annotation tools in educational settings. In order to make the number of the reviewed articles manageable but comprehensive, exhaustive review with selected citation approach (Cooper, 1988) was implemented. Specifically, the studies were included in the literature review if they met each of the following criteria:

1. The study employed an online SA tool that embodied at least the following three features: (a) annotations, (b) highlights, and (c) information sharing/collaboration online platform.
2. The study examined either (a) usability of SA technology in educational settings or (b) effects of SA technology on general learning benefits or learning outcomes.
3. The study focused on higher education audience. Studies that employed K-12 population were excluded.
4. The study reported research methods (i.e., participants, instructional interventions, instruments, research design, etc.) and results. Studies that employed quantitative research method reported statistical analysis methods and results.
5. The research presented in one study did not overlap research from another study.
6. The article was written in English.

The search was conducted using various search terms and keywords such as *online collaboration tools*, *annotation tools*, *online team learning tools*, *learning outcomes*, and *achievement*. Since the SA tools are an emerging technology that has not yet been adopted widely, the search was not limited to a particular date range, research methods, specific learning outcomes, or publication type. From over 90 initially collected articles, eight experimental and quasi-experimental studies and eight evaluation/survey studies met the inclusion criteria for this literature review.

3. Findings

Emerging educational technology opens a new horizon of instructional design and learning effectiveness issues associated with embedding the technology into learning activities. Researchers need, first, to explore the tool(s) and examine its usability in educational settings. Next step would be using this new body of knowledge to investigate the effectiveness of the tool in educational settings.

This section begins by presenting evaluation/survey studies aiming at evaluating students' attitudes toward SA technology, their perceptions toward using SA in educational settings, and ways to improve various SA-based learning activities. Following the evaluation/survey studies, a more rigorous experimental and quasi-experimental research focusing on effects of SA and uses of SA in educational settings is presented. Each study is discussed in terms of research purposes, SA tool,

instructional strategies, instruments, outcome measures, research design, participants, and results. Table 2 summarizes the reviewed SA studies.

3.1. Evaluation/survey studies

This section presents eight empirical evaluation/survey studies that examined various SA tools, their usability in educational settings, instructional design issues associated with embedding the tool into learning activities, and students' attitudes toward the tool.

3.1.1. Use of annotations for information search tasks and reading comprehension

In order to understand how to better support active reading and annotations in the digital context, Kawase et al. (2009) carried out a comparative study of online and paper-based annotations. The purpose of their research was to investigate the types of annotations encountered online and on paper, and to find differences between the two environments. Specific attention was given to the type of annotations, their function, and perceived difficulties in creating and using these annotations.

To explore how students annotate electronic resources, the researchers have developed SpreadCrumbs, an annotation system that provides an interface for adding post-it notes and crumbs (i.e., personal reminders) to any point within a Web page. In addition, SpreadCrumbs users can bookmark any web-resources and share their annotations and crumbs with others.

The researchers conducted two experiments. The purpose of the first experiment was to explore how students annotate an electronic text; while in the second experiment, the researchers wanted to understand students' paper-based annotation behavior in order to inform design of the SpreadCrumbs SA systems. Eighteen participants, who rated themselves as proficient in working with computer and internet technology, participated in the first experiment, a laboratory study. During the experiment, the participants were given information search task, where they created 207 annotations, covering 81 different Web resources. None of the users demonstrated problems regarding the use of the tool. The participants demonstrated enjoyment with the tool interface and functionalities. After the short introduction, all of the participants performed the tasks of annotating and consulting annotated resources without any effort or mistake.

Twenty-two PhD students and post-docs participated in the second experiment. A group of two or more students collaboratively annotated 9% of the documents. All except two participants shared their comments via email or other online communication tools. Two participants worked together on the same sheet of paper that contained annotations from both students. The participants were asked to annotate three articles or papers that they read. The researchers found 1778 annotations and an average of 3.08 annotations per page. The far majority of the annotations (73%) involved the highlighting of text. The authors also noticed that all of the participants annotated shared documents more carefully than the unshared ones.

To examine in more details the annotation strategies, the researchers asked the participants to classify the goal of reading the paper. The authors distinguished between the following categories: *reading for writing* (i.e., common activity of reading related articles to extract ideas and references), *reading for learning* (i.e., act of being updated in some particular field), *reviewing* (i.e., reading papers to give feedback to the author) and *other*. In summary of the observations, the authors identified two main clusters of annotations from the paper-based experiment: (a) relevance adjustment annotations where implicit highlight and signaling indicated different levels of importance in the text and (b) contributive annotations where explicit readable remarks were attached to the text.

The authors concluded that the higher amount of annotations per page showed that these annotations affected memorizing parts of the text by actively engaging the learner in the annotation process.

Table 2
Studies' characteristics.

Study	Research method	Sample size (n)	Topical domain	Assignment and Intervention	SA tool and feature(s)	Outcome measured	Results
Archibald (2010)	Experimental	128	English: Reading and persuasion course	Reading comprehension; 8 treatments: various combinations of SA activities, whole task instructional strategy, and collaboration strategy	HyLighter: annotation, highlights, and information sharing	• Reading comprehension • Critical thinking • Meta-cognitive skills	• Initial performance during the instructional activities was decreased followed by an increase in performance in the one month delayed tests
Gao and Johnson (in press)	Survey	33	Educational technology course	Reading comprehension: read and annotate an online article, share annotations with classmates, and respond to classmates' annotations	Diigo: annotation, highlights, and information sharing	• Students' perceptions on using Diigo	• Diigo was found quite supportive • SA tools in learning settings might be distracting
Hwang, Wang, and Sharples (2007)	Quasi-experimental	70	Multimedia application course	2 Treatments: 2) Non-SA (control) 1) SA-based activities	VPen	• Quantity of annotation • Learning achievements • Attitudes toward VPen system	• Learning achievements of experimental group were significantly higher than the non-experimental • Annotation increased learners' motivation and interaction
Johnson, Archibald and Tenenbaum (2010) – Study 1	Quasi-experimental	254	English: Reading and persuasion	Reading comprehension; 5 treatments: various combinations of highlighting and annotating + no SA tool (control group)	HyLighter: annotation, highlights, and information sharing	• Reading comprehension • Critical thinking • Meta-cognitive skills	• No significant difference on the three outcome variables
Johnson, Archibald and Tenenbaum (2010) – Study 2	Quasi-experimental	267	English: Reading and persuasion course	Reading comprehension; 4 Treatments: 1) Group SA activity + Group comparison 2) Individual SA activity + Group comparison 3) Individual SA activity + Individual comparison 4) Individual SA activity + Individual comparison	HyLighter: annotation, highlights, and information sharing	• Reading comprehension • Critical thinking • Meta-cognitive skills	• Students achieve better outcomes on measures of reading comprehension and meta-cognitive skill, but not critical thinking, when SA activities include small team collaborations
Kawasaki, Sasaki, Yamaguchi and Yamaguchi (2008)	Experimental	20	University level courses – unspecified	Text reading on computer monitor; 2 Treatments: 1) Experimental group (n = 10) read text with highlighting provided, 2) Control group (n = 10) read text without highlighting	Highlights	• Reading comprehension	• Experimental group scored higher than control group on multiple-choice questions but not on fill-in-the-blank questions
Kawase, Herder, and Nejd (2009)	Experimental	30	University level courses – unspecified	Information search task; 3 Treatments: 1) Internet search engine 2) Delicious social bookmarking 3) SpreadCrumbs annotation	SpreadCrumbs: annotations and highlights	• Learners' reading behavior • Time spent on information search • Search strategies	• Annotation group found information significantly faster than the other 2 groups
Kawase, Herder and Nejd (2009)	Comparative study (2 experiments)	Study1: 18 Study 2: 22	University level courses – unspecified	Study 1: Information search task Study 2: Reading and annotating and article	SpreadCrumbs: annotations and highlights	• Quality of annotations • Attitudes toward the SA system	• The participants demonstrated enjoyment with the tool interface and functionalities
Mendenhall and Author (2010)	Qualitative (interviews)	6	English: Reading and persuasion course	Reading comprehension: peer-critique process	HyLighter: annotation, highlights, and information sharing	• Attitudes toward SA	• SA tool offered an appropriate environment for peer critique activity

Table 2 (continued)

Study	Research method	Sample size (n)	Topical domain	Assignment and Intervention	SA tool and feature(s)	Outcome measured	Results
Nokelainen et al. (2003)	Survey	31	Statistics	Worked collaboratively in pairs on annotating, highlighting, and discussing an online document	EDUCOSM system	• Annotations quality • Cognitive learning outcomes in statistics	• Learners, who were willing to do real work with the SA tool, produced both highest quality annotations and learning outcomes
Nokelainen et al. (2005)	Survey	50	Graduate hypermedia course	Worked collaboratively in pairs on annotating, highlighting, and discussing an online document	EDUCOSM	• Motivation • Usability of the SA tool • Learning achievement	• The level of motivation had a positive effect on instructional activity in the system and the final grade • The learners, who reported to have good time management strategies, were the most active users of the system
Razon, Turner, Johnson, Aarsal, and Tenenbaum (submitted for publication) Study 1	Experimental	27	Education: Classroom assessment undergraduate course	Reading comprehension; 3 Treatments: 1) HyLighter 2) Non-HyLighter 3) Non-HyLighter	HyLighter: annotation, highlights, and information sharing	• Reading comprehension • Emotions and motivation	• Non-significant on any of the outcome variables • Descriptively, HyLighter SA treatment performed better than the non-HyLighter SA treatments
Razon, Turner, Johnson, Aarsal, and Tenenbaum (submitted for publication)	Experimental	40	Education: Classroom Assessment graduate course	Reading comprehension; 2 Treatments: 1) HyLighter 2) Non-HyLighter	HyLighter: annotation, highlights, and information sharing	• Reading comprehension • Emotions and motivation	• Non-significant on any of the outcome variables
Samuel, Kim, and Author (in press)	Comparative study (survey)	20	Sports psychology	Reading comprehension: annotating and highlighting an article	HyLighter: annotation, highlights, and information sharing	• Usability of SA tool • Motivation	• High positive attitudes toward the usability of HyLighter • Positive attitudes toward collaborative work experience using HyLighter

According to the authors, the high amount of highlighting/markings helped to reduce cognitive overload due to switching of tasks and to keep focused on the main task (reading) while still providing meaningful cues.

Kawase et al. (2009) argue that the results of the two studies along with the existing research on paper-based annotations have revealed that in paper-based environment annotations clearly support the learning process. However, in online environment, the act of annotations has become a burden due to several issues. First, the SA tools are not suitable enough to provide an effortless environment for annotating. Second, most of the students avoid reading from a screen and interacting with a text via keyboard and mouse as opposed to paper-based annotations where the interaction is via pen or sticky notes. A comparison of online and paper-based annotations revealed that online annotations were short and served the purpose of re-finding, sharing, or commenting. In paper-based documents, the high amount of highlight and annotations had an intrinsic value.

Nevertheless, it is unclear whether the results of the comparison between the two studies can be generalized. First, Kawase et al. (2009) have varied instructional tasks and participants' background characteristics across the two studies, which make the interpretation of the results differences between the two studies quite complicated. Second, the employed sample sizes were very small.

3.1.2. Peer critique SA activities

In addition to investigating differences between paper-based and online annotations, a number of studies were conducted to examine ways to improve various SA-based learning activities, students' attitudes toward SA, and their perceptions toward using SA in educational settings. Mendenhall and Author (2010) conducted a formative evaluation using basic qualitative research methods including interviews to determine student's perceptions on the benefits and weaknesses of using SA technology for peer critique instructional activity.

A simulation of a proposal critique process was implemented through a collaborative work of pairs of students in the HyLighter SA system. HyLighter is an online SA system that allows user to highlight and annotate in an electronic text and share it with others.

In each pair, one of the students (author) provided a proposal to a second student (peer reviewer), who would then review and critique the proposal by responding to evaluation questions that were typed into HyLighter as a general note (one that is not linked to a specific portion of the text) allowing the peer reviewer to answer the evaluation questions while having direct access to the document. The presented questions often included a line-by-line evaluation of topic sentences, transition words, descriptions of independent and dependent variables, and other grammatical and conceptual components of a research proposal. Students (peer reviewers) annotated areas that needed to be improved by inserting their comments using HyLighter annotation features. What was unique about using HyLighter for this type of learning activities was that this learning system allowed both students (author and peer reviewer) not only to observe the final proposal critique, but being an integral part of the proposal critique process.

Six participants (one peer reviewer and five student authors) participated in this study. Overall, the participants felt that the SA tool offered an appropriate learning environment for peer critique activity. According to students' feedback, peer-critique activity implemented in HyLighter was particularly useful for receiving a feedback from their peers. HyLighter annotations and annotations tagging features allowed students to see all comments tied to relevant parts of the proposal.

Based on the results of this formative evaluation study the authors suggested that teams of 2–3 people use SA system to answer the evaluation questions, critique the research proposal, and discuss the critique in class. The major benefit of having groups of peers work on the same document instead of individual peers is when students study in teams it reduces the redundancy of critique items. In this way, the class discussion can focus on fewer, but more in-depth issues.

3.1.3. Critical analysis and comprehension

Samuel, Kim, and Author (in press) conducted a comparative study design employing a survey to evaluate usability of students' experiences using HyLighter. Students worked in groups to compare and contrast two philosophies in sports psychology. They read an article and then answered reading comprehension questions to assess their overall knowledge of the content in the articles. The major purpose of the reading assignment was to assess students' critical analysis and comprehension skills. The participants were 20 students enrolled in a sport psychology course. Students highlighted sections of the article that supported their answers and provided their personal opinion reflecting on the reading. The class assignment did not involve sharing individual or class responses until each one submitted their answers. In this way, the instructor could (1) capture a baseline score of individuals and groups, and (2) prevent answer sharing between the groups. In addition, this prevented students' opinions to be influenced by others' responses. After all the annotations and responses were submitted, the instructor chose two model responses of opposing points of view about the assigned article shared them with the students using the SA system. The students were asked to respond to these postings within the SA system, which allowed them to see and react to a variety of opinions.

The usability of HyLighter was assessed on a five-point Likert scale (1 = negative attitudes and 5 = positive attitudes). The authors found that students had high positive attitudes toward the usability of HyLighter (Mean = 4.37, SD = 0.78, $n = 20$). Students were satisfied with HyLighter's interface, ease of navigation, and operation of the tool. Students also reported positive attitudes toward their collaborative work experience using HyLighter (Mean = 4.62, SD = 0.51, $n = 9$).

3.1.4. Online SA-supported collaborative learning

Several studies were conducted to explore how various SA tools (i.e., Diigo and EDUCOSM) could be used to support online cooperative learning. Nokelainen et al. (2003) examined the usefulness of a shared document-based annotation tool, EDUCOSM, in real-life collaborative learning situations (i.e., learners take responsibility for their own learning). The EDUCOSM system consists of a set of tools including search, filters, and others for asynchronous collaborative knowledge construction. The authors also investigated how learner's self-rated use of learning strategies was related to cognitive learning outcomes (final examination of the course) and completion of various tasks in the system (i.e., online group formation and peer-to-peer annotation of the course material).

Thirty-one vocational education in-service teachers enrolled in a web-based university-level statistics course participated in this study. The learners engaged for two weeks in online distance learning in the SA system. During this time, the students anonymously formed a group of two and then worked in pairs on annotating, highlighting, and discussing an online document.

The authors used a questionnaire to measure learning strategies. The questionnaire (16 items) consisted of three dimensions of professional learning: motivation, learning strategies and social abilities. Final examination measuring subject-related (i.e. statistical topics) cognitive outcomes was conducted at the end of the course. After the course, the course lecturer evaluated the quality of annotations made by the students in the EDUCOSM system individually on a scale from 0 to 3.

The results with 74% classification accuracy showed that the quality of annotations was rated higher for male than for female students. Results indicated that auditory and verbally oriented students, who liked to have practical training from teacher and liked to ask questions, generated lower quality annotations compared to those who spent the most time in the system. With respect to the final examination cognitive outcome, the results with 63% accuracy showed that auditory and verbally oriented students, who needed teacher's feedback and liked to learn from practical training, scored lower on the final examination.

In general, the overall results indicated that those learners who were willing to do real work with the tools provided by the system, were able to elaborate what they were doing, and produce both highest quality annotations and learning outcomes.

To further evaluate how the EDUCOSM SA tool could be embedded into instructional activities, Nokelainen, Miettinen, Kurhila, Floréen, and Tirri (2005) conducted another study. The authors aimed at investigating the effects of using EDUCOSM SA tool on students' motivation, learning strategies, and social ability. The purpose of this study was to empirically investigate the usefulness of the SA tool in two different real-life web-based university-level courses: 27 Finnish vocational education in-service teachers enrolled in a web-based post-graduate level statistics course (i.e., adult learners) and 23 Finnish masters students enrolled in a web-based adaptive hypermedia course (i.e., adolescent learners). The authors used different samples for two reasons. First, they were interested in whether SA tool serves for both masters and post-graduate students. Second, the authors examined whether there is a difference between masters and post-graduate students' satisfaction with the SA system. The SA-based instructional activities in the both web-based courses were similar to the ones from the Nokelainen et al. (2003) first study. The learners worked in pairs for two weeks on annotating, highlighting, and discussing an online document.

The study design consisted of three data collection phases: 1) a pretest measuring self-rated motivation, learning strategies, and social ability, 2) log file data showing actual use of the system features, and 3) a posttest in a form of an email survey.

The results for both groups showed that the level of motivation had a positive effect on instructional activity in the system and the final grade. The learners, who reported to have good time management strategies, were the most active users of the system. The level of social ability predicted both number of consecutive comments in documents and threads in document-related newsgroup discussions. Log file data analysis showed that user activity in the system was positively related to the final grade in both samples. Results also revealed that self-rated level of motivation had a positive effect on final grade. Students' self-rated need for receiving performance-related feedback from the teacher or tutor correlated negatively with observed activity in the system. High meta-cognitive abilities correlated positively with the final grade. Additionally, authors found evidence that study success was positively related to the learner's activity in the system. Those students who found it most rewarding to research a subject as thoroughly as possible were the most active annotators in the course. Quality of annotations in the system correlated positively with activity in the system and final grade.

Results of the post survey showed favorable attitude toward the system. Respondents from the both web-based courses reported that the system brought benefit to the learning process, changed their studying habits favorably, and that they would use the system in other courses.

This study design posed a number of challenges in interpreting differences among environments since Nokelainen et al. (2005) varied various experimental conditions (i.e., participants age, education level, and study domain) across the two different web-based courses. Nevertheless, it allowed making stronger judgments regarding the generalizability of the findings, which is one of the major strength of this study. Since the level of motivation had a positive effect on activity in the system and the final grade across the two web-based courses, it is possible to assume robustness of this finding.

In addition to EDUCOSM, a Web 2.0 SA tool, Diigo, was examined and tested out to explore whether and how this tool can support online collaborative learning (Gao & Johnson, in press). Thirty-three pre-service teachers enrolled in an undergraduate educational technology course participated in the survey study. Students were instructed to read and annotate an online article, share annotations with classmates, and respond to classmates' annotations using Diigo while focusing on two questions suggested by the instructor. The whole activity lasted for one week. At the end of the week, students

completed a five-point Likert scale survey designed to measure students' perceptions on their experience participating in the activity and students' online annotations were collected. Overall, students perceived using the SA tool in learning as somewhat (54.5%) or quite (33.3%) supportive. Nevertheless, some disadvantages of using Diigo were reported as well. Among the students' concerns posed by Diigo SA technology were challenges with (a) navigating multiple comments on a webpage, and (b) having a holistic group collaboration effort. In addition, students reported that other students' comments and annotations were very often more interesting than the actual article. Thus, using SA tools in learning settings might be somewhat distracting.

3.2. Experimental and quasi-experimental studies

This section presents eight experimental and quasi-experimental studies evaluating the instructional benefits and uses of social annotation technology in higher education. The studies are organized based on learning outcomes and instructional roles associated with the SA.

3.2.1. Reading comprehension, critical thinking, and meta-cognitive skills

A series of experimental and quasi-experimental studies were conducted to examine the effects of SA on reading comprehension, critical thinking, and meta-cognitive skills. Johnson, Archibald and Tenenbaum (2010) conducted two quasi-experimental studies evaluating the effects of various combinations of highlighting and annotating instructional activities combined with review of peers' and instructor's comments using HyLighter SA tool on reading comprehension, critical thinking, and meta-cognitive skills.

In the first study, 254 freshman students enrolled in English course that focused on argument and persuasion, were assigned into five treatments, and were asked to practice one of the various learning activities. The first four treatment conditions involved HyLighter-supported instructional activities, while the fifth treatment (control) group read a printed text without using HyLighter (see Table 3).

To measure reading comprehension, critical thinking, and meta-cognitive skills learning outcomes, the researchers developed an instrument for each of the outcomes (three instruments in total). The Reading-comprehension skills instrument comprised of three true/false reading comprehension questions. The Critical-thinking skills instrument consisted of two open-ended questions aimed at assessing critical thinking skills. The Meta-cognitive skills instrument constituted of two open-ended questions probing students' meta-cognitive skills such as providing the rationale or reasons for the reached conclusion. No significant difference on the three outcome variables, i.e., reading comprehension, critical thinking, and meta-cognitive skills, was revealed between the four instructional interventions that used HyLighter and the control group instruction omitting HyLighter, thus showing that instructional activities incorporating HyLighter SA did not result in better learning outcomes in this study.

Likewise, the second study presented in Johnson et al. (2010) evaluated the effect of HyLighter SA instructional interventions on reading comprehension, critical thinking, and meta-cognitive skills in English course.

However, the emphasis was on comparing the effects of individual versus collaborative learning in the SA learning environment. A two-step instructional activity was designed to examine the effects of social annotation instructional practices on reading comprehension, critical thinking, and meta-cognitive skills in either collaborative or individual learning settings. In both steps, 267 freshman students enrolled in English course that focused on argument and persuasion worked either collaboratively in small groups or individually on course activities using HyLighter. During the first step, students either collaboratively or individually (a) read an article presented in HyLighter, (b) highlighted and annotated the article to better understand the text, (c) answered reading comprehension questions, (d) "chunked" the article (i.e., the process of breaking down a text into smaller pieces in order to make sense in relation to the big picture), and (5) identified the article's thesis (i.e., identified a section of the article as a statement of the author's thesis by highlighting and annotating the section). In the second step, students either collaboratively or individually compared their responses to those of the instructor. This study design generated four different conditions to evaluate their effect on learning outcomes. The instruments from the first Johnson's et al. (2010) study were administered in this study as well.

The results of the second study showed that the students, who were collaboratively engaged in at least one or both steps, demonstrated better meta-cognitive skills as compared to individual (non-collaborative) learning in both steps. This effect was shown/supported in similar settings as noted in previous research (Kirschner, Paas, & Kirschner, 2009; Laughlin, Bonner, & Miner, 2006). Another finding demonstrated that when the students completed either one or both of the annotation and/or comparison tasks collaboratively, they benefited in both reading comprehension and meta-cognition. No significant differences between groups were found for critical thinking outcomes. The results of this study revealed that the instructional strategies that lacked the "social" (collaborative) SA feature were not more effective than reading a textbook.

According to the authors, there were two confounding effects, which could hinder the SA instructional effect: (1) insufficient exposure time to the intervention, and (2) limited time to learn how to use HyLighter. Nevertheless, the two studies provided important evidence regarding the effects of SA-based instructional activities on development of various cognitive skills.

Another experimental study was conducted in a laboratory setting to examine various instructional strategies aimed at improving students' reading comprehension, critical thinking, and meta-cognition when using the HyLighter SA tool (Archibald, 2010). The author designed and evaluated instructional solutions of using social annotation technology in higher education settings. In order to get the most learning benefits from using SA technology, Archibald (2010) suggested to combine three instructional strategies: (a) social annotation strategy, (b) First Principles of Instruction or whole task strategy (Merrill, 2002), and (c) collaboration strategy (Johnson, Khalil, & Spector, 2008). The integration of these three instructional strategies created an environment where students could participate collaboratively in an instructional activity designed in line with First Principles of Instruction using SA technology. In order to explore the effects of various combinations of the three instructional strategies (i.e., SA strategy, First Principles of Instruction, and collaboration strategy) and SA tool on students learning, eight experimental conditions were designed (Table 4). One hundred twenty eight freshmen students enrolled in English course were randomly assigned into one of the eight treatments.

The author developed three instruments for examining each of the three cognitive skills, i.e., reading comprehension, critical thinking, and meta-cognition. A Subject Matter Expert (SME) validated the instruments, which were subsequently reviewed and edited by three additional experts in the field. However, no reliability measures were reported.

Both positive and negative significant effects of the employed instructional strategies on learning were observed. Specifically, looking

Table 3

Instructional methods used in Johnson, Archibald and Tenenbaum (2010) Study 1. Adapted from "Individual and team annotation effects on students' reading comprehension, critical thinking, and meta-cognitive skills", by T. E. Author, T. N. Archibald, G. Tenenbaum, 2010, *Computers in Human Behavior*, 26(6), pp. 1496–1507.

Method	Description
1	Highlight and annotate (H&A)
2	H&A + review of peer highlights and annotations
3	H&A + review of lead instructor highlights and annotations
4	Preview lead instructor highlights + H&A + self-reflection (i.e., review and compare to expert highlights and annotations but no review of peer input)
5	Read hard copy of article, no use of HyLighter (control)

Table 4

Instructional methods used in Archibald (2010).

Adapted from "The effect of the integration of social annotation technology, first principles of instruction, and team-based learning on students' reading comprehension, critical thinking, and meta-cognitive skills", by T. N. Archibald, 2010, Dissertation, Florida State University, Tallahassee, FL.

HL				No HL			
Group		Individual		Group		Individual	
MIA/NMIA	NMIA/MIA	MIA/NMIA	NMIA/MIA	MIA/NMIA	NMIA/MIA	MIA/NMIA	NMIA/MIA
1	2	3	4	5	6	7	8

at the social annotation strategy effect, students experienced extraneous cognitive load that negatively affected their initial performance because of instructional activities. However, there was a positive effect on students' achievement in the one month delayed tests as compared to the non-social annotation strategy treatment. One of the major limitations of the study was the duration of the instructional intervention, which lasted in total up to 4 h. Most likely, the instructional interventions should be longer in order to more accurately evaluate long-term results of the interventions. These results are similar to Johnson's et al. (2010) study, wherein instructional interventions lasted for a total of about 8 h. Despite these limitations, this study produced promising results showing effects of the use of the SA technology on students' reading comprehension, critical thinking, and meta-cognitive skills.

3.2.2. Reading comprehension and emotions and motivation toward reading task

Razon, Turner, Johnson, Aarsal, and Tenenbaum (submitted for publication) aimed at evaluating the effect of different SA practices on (a) reading comprehension and (b) emotions and motivation. The researchers conducted two experimental studies using Highlighter SA tool in undergraduate (Study 1) and graduate (Study 2) course settings. In both studies, students were instructed to (1) highlight and annotate selected articles, and (2) highlight and annotate selected articles and review peer highlights and annotations. In each study, students were assigned to two treatments: (1) HyLighter, and (2) non-HyLighter. In the HyLighter treatment, students used the social annotation tool (e.g., HyLighter). Students from the non-HyLighter treatment read a printed hard copy of the article without using HyLighter, and interacted with peers on the article without receiving guidelines from the instructor.

The major differences between the two studies were in research design. In the first study, 27 undergraduate students enrolled in a southeastern university Classroom Assessment course that focused on measurement and evaluation of academic achievement, were assigned to three sections. These sections were taught by different instructors but using the same instructional materials and textbook. Students in one of the three sections used social annotation tool (e.g., HyLighter), while students in the remaining two sections read a printed hard copy of an article without using social annotation tool.

The authors developed a number of instruments including a Reading Comprehension quiz and a Reading Affects Questionnaire for the first study. After reading each assigned articles students completed the Reading Comprehension quiz consisted of multiple-choice items aimed at assessing reading comprehension skills. The items were developed by the course lead instructor and validated by a team of three experts in the field. Students' emotions and motivation related to reading assignments and academic success were assessed using the Reading Affects Questionnaire. This questionnaire included seven self-report adjectives aimed at assessing students' affect toward reading, annotating, and interacting with other peer readers on the articles.

In the second study, 40 students enrolled in a southeastern university graduate Classroom Assessment course that focused on measurement and evaluation of academic achievement were assigned to two treatment conditions within the class where two-phase

experiment was carried out. During the first phase, group 1 used HyLighter and group 2 did not. Intermittently in the second phase, group 1 did not use HyLighter while group 2 did.

A set of instruments was developed for the second study. Students' reading comprehension was assessed using a Reading-Comprehension Quizzes consisted of 10 multiple-choice questions. Students' achievement goal orientations were assessed using the Achievement Goal Questionnaire-Revised (AGQ-R; Elliot & Murayama, 2008, as cited in Razon et al., submitted for publication). The AGQ-R includes 12 items aimed at assessing students' achievement goals within the hierarchical model of the approach-avoidance mastery/performance achievement goals (Elliot, 1997, as cited in Razon et al., submitted for publication). Students' test anxiety, self-efficacy, and control of learning were assessed using the Motivation Strategies for Learning Questionnaire (MSLQ; Pintrich, Smith, García, & McKeachie, 1991, 1993, as cited in Razon et al., submitted for publication). The MSLQ includes 81 self-report items aiming at assessing the nature of motivation and use of learning strategies. The reliability measures of the AGQ-R and MSLQ were moderate to high.

In both studies, the mean differences were not statistically significant on any of the outcome variables. However, descriptive analysis of means from the first study showed that the HyLighter treatment performed better than the non-HyLighter treatments. These results are consistent with previous findings indicating positive effects associated with improved critical thinking and meta-cognitive skills in task-specific assignments (Johnson et al., 2010). In addition, descriptive differences in motivation for reading and positive emotions in favor of HyLighter treatment were found. The authors suggested that employing larger sample sizes might result in significant differences between the HyLighter and non-HyLighter treatments. Such trends were not observed in graduate students. The authors attributed the differences between the two studies to the following. In the first study, an instructor provided a constant assistance on how to use HyLighter; while in the second study, students did not receive any form of support. According to authors, these differences strengthen the notion of the importance of providing guidelines for using even simple educational technology tools (Jeong & Hmelo-Silver, 2010). However, the Hawthorne effect might explain these differences as well. The Hawthorne effect refers to the fact that participants tend to perform better due to attention from the research team and not because of changes in the instructional interventions.

3.2.3. Reading comprehension

Another study examined the influence of highlighting as a memory prompt in text reading (i.e., reading comprehension) on a computer monitor (Kawasaki, Sasaki, Yamaguchi, & Yamaguchi, 2008). The participants in this study were 20 college students in a University in Japan. Kawasaki et al. (2008) randomly assigned the students into two treatment conditions. The participants in the experimental group ($n = 10$) read the highlighted text provided on a computer monitor, and those in the control group ($n = 10$) read the text without highlights. The researchers prepared two types of study materials on the web site, one for the experimental group and the other for the control group. For

the experimental group, important words and phrases in the text were emphasized with highlighting by red and pink. The text for the control group was the same as the one for the experimental group except for no highlighting. After interacting with the text, students from the both groups took a reading comprehension posttest. The posttest included two sections (section 1 and 2). Section 1 included multiple choice questions and section 2 included fill in the blank questions.

Kawasaki et al. (2008) hypothesized that learners' comprehension in reading texts on a computer monitor would be higher when important words, phrases, and sentences were pre-highlighted than when they were not highlighted. Although the sample size was very small, results revealed that the experimental group scored significantly higher than the control group in section 1, but not in section 2. The authors concluded that prompts had potential in enhancing learning on a computer monitor. Therefore, they recommended that teachers should pre-highlight important words, phrases, and sentences in the text of the learning materials when students are using a computer.

3.2.4. Learning achievement, attitudes, and relationships between quality of annotations and learners' characteristics

Hwang, Wang, and Sharples (2007) conducted a quasi-experimental study to explore the usage of the Web-based annotation tool and to investigate its influence on online learning. The authors designed a Web-based tool, called VPen, for creating and sharing annotations. The independent variables of this research were: (a) field independent learners (FI)/field dependent learners (FD) cognitive styles and (b) learning scenarios. The dependent variables were (a) quantity of annotation and (b) learning achievements. The research questions in this study were the following: (1) What are the students' perceived attitudes toward VPen system after their usage? (2) What are the effects of different cognitive styles on annotation? (3) What are the effects of different annotation sharing scenarios on quantity of annotation and its influence toward learning achievements? (4) What is the relationship between quantity of annotation and learning achievements? Is there a significant difference in quantity of annotation between high achievement students and low achievement ones?

A quasi-experimental method was used in this research. Two intact classes ($n=70$) of freshmen college students in a Multimedia Applications course participated in the experiment that took place over 4 months. During this period, students from the experimental class ($n=38$) participated in three learning consequent activities where they annotated web-based materials using the SA tool (1) individually, (2) group annotation sharing, and (3) full annotation sharing. Students from the non-experimental class ($n=32$) did not use SA tool. They were reading online learning materials individually during the same period.

Data was collected using a survey consisting of four subscales: perceived usefulness, perceived ease of use, learning satisfaction, and the application of the annotation system in practical learning scenario. This data showed that with respect to perceived usefulness and simplicity, most of the students thought that the annotation system did improve their online reading performance. They also perceived that it was easy to use the annotation system. With respect to perceived satisfaction of learning with the annotation system, most of the students thought that the materials with annotation system increased their interest, happiness, and achievements in learning. Additionally, they thought the annotation system did improve interaction between learner and content of materials. The students had positive attitudes toward the application of the annotation system to learning.

Moreover, results revealed that the learning achievements of the experiment class that used SA tool were significantly higher than that of the control class. That is, the use of VPen annotation system appeared to help students' learning achievements in the annotation sharing scenario group. One limitation of this study was that the web tool restricted the students' abilities to fully annotate the learning materials. Despite

this limitation, the authors concluded that there was value in further study of collaborative learning through shared annotation.

3.2.5. Information search

In addition to evaluating SA learning effectiveness, the literature review has shown that SA technology has been used for information search tasks (Kawase et al., 2009). The authors examined the possible benefits of annotations over bookmarks web-based tools. Specifically, the authors used Delicious social bookmarking service and SpreadCrumbs web-based annotation tool to understand users' annotation behaviors when reading, and to identify the benefits and drawbacks of SA tools. SpreadCrumbs is an online annotation tool that allows the users to place annotations within Web resources, either for themselves or for other users.

In their experimental study, the researchers randomly assigned 30 adult participants into three equivalent treatment groups ($n=10$ per group): (1) Internet search engine, (2) Delicious social bookmarking, and (3) SpreadCrumbs annotation. Then they compared the time the participants spent on relocating the information that had been previously found using annotations, bookmarks, or Internet search engine. Even though the group sizes were small, the authors found that the annotation group found the needed information significantly faster than the bookmarking and the search engine groups. No differences were found between the bookmarking and the Internet search engine strategies. The authors suggested that these findings demonstrated a need for annotation tools and the benefits that SA tools could offer. Kawase et al. (2009) claimed that SA technology could improve users' effectiveness and efficiency in information search tasks.

4. Discussion

Out of more than 90 identified papers related to SA technology, only 16 experimental studies met the inclusion criteria and, therefore, were included in the present literature review. Among the included research studies, were eight experimental or quasi-experimental studies and eight evaluation/survey studies. SA technology was used to enhance and facilitate learning in various domains of study such as English (e.g., Archibald, 2010; Johnson et al., 2010), classroom assessment (Razon et al., submitted for publication), statistics (Nokelainen et al., 2003) and multimedia (Hwang, Wang, & Sharples, 2007).

The literature review has revealed empirical findings addressing the research questions pursued in this paper. First, empirical research suggests that SA tools lead to learning gains in higher education. Although the empirical research in the area of SA technology is extremely limited, the literature provides some promising and supporting evidence regarding the potential effectiveness of integrating social annotation tools into learning. Several benefits related to the use of SA tools in education setting have been identified based on the review of the eight experimental and quasi-experimental studies. A comparison of students engaged in SA learning activities versus those that did not, has revealed that SA-based learning activities contribute to improved critical thinking, meta-cognitive skills, and reading comprehension (Razon et al., submitted for publication; Johnson et al., 2010; Hwang et al., 2007). In addition, preliminary findings have shown that SA technology promotes motivation for reading and contributes to higher frequency of positive emotions and lower frequency of negative emotions (Razon et al., submitted for publication).

Notwithstanding the benefits, there is an initial performance cost associated with using social annotation technology in education. Archibald (2010) found that most of the highlighting and annotating instructional strategies had initial negative effects (pre to post) on students' achievement. However, students' one month delayed performance scores (post to delayed) superseded the pre-test scores where as the control group delayed performance scores dropped more dramatically and were significantly lower than the treatment group. The author attributed the initial performance decrease to the

elevated extraneous cognitive load that students experienced while working with the technology on specific learning tasks.

In addition to the experimental and quasi-experimental studies, eight evaluation/case studies were included in the literature review. These studies focused on assessing students' attitudes toward using SA tools in higher education. The findings from the evaluation/case studies provided empirical evidence regarding the general learning benefits and learning outcomes linked to SA. Overall, it appeared that students liked using social annotation technology and felt that this type of technology facilitated and supported learning (e.g., Gao & Johnson, *in press*; Nokelainen et al., 2003; Samuel et al., *in press*). Researchers found that the quality and amount of annotations had a significant correlation with student performance (Kawase, Herder, & Nejd, 2010). Moreover, the most active SA students had good time management strategies and students' social ability was positively related to the amount of annotations and final grade (Nokelainen et al., 2005). It appeared that there was a strong correlation between strong time management and social ability skills and one's ability to engage in social annotating. However, some students reported that using of SA tools in learning settings could be somewhat distracting (Gao & Johnson, *in press*), since the interactions themselves might be much more interesting than the instructional activity.

The last research question pursued in the paper focuses on how SA technology has been embedded into instructional activities and what types of learning have been facilitated the most through using SA technology. The literature review has revealed that SA technology can serve different roles in different instructional tasks. The majority of existing research investigated how SA features such as annotations, highlights, and information sharing could be embedded into instructional activities to promote reading comprehension, critical analysis, and meta-cognitive skills, and foster communication between students and peers, and students and instructors. Various instructional strategies incorporating SA technology were implemented. For instance, students either worked collaboratively or individually on highlighting and annotating a paper, and then compared their work with peers' or lead instructor's annotations (e.g., Johnson et al., 2010). SA tools were also employed to support peer critique activities (Mendenhall & Johnson, 2010). In general, the literature review suggests that SA technology can be an effective tool for supporting and enhancing various reading comprehension activities. Using SA tools, students can review and analyze reading materials, bookmark and highlight important information in text, review peers' and instructor's comments, and provide feedback to peers.

In addition, SA tools were used for information search tasks (Kawase et al., 2009), where students initially tagged specific places in an electronic text, and later on used these tags to find desired information in the text. This technique could be very useful for working with long articles. In addition, SA technology was used for bookmarking electronic resources and sharing them with others.

5. Recommendations for using social annotation technology

Results of the gathered studies provided practical suggestions on (1) effective use of SA technology in educational settings and (2) SA tools design implications. Many researchers emphasized the importance of providing adequate technology training for teachers and students before actual implementation of SA technology-supported instructional activities in class (e.g., Razon et al., *submitted for publication*; Johnson et al., 2010; Bateman, Brooks, McCalla, & Brusilovsky, 2007). When considering SA-based instructional activities, students should be provided with sufficient time to get used to work with the new technology (at least 4 h) in addition to receiving training on how to use the technology (Archibald, 2010). It is also very important to help teachers become aware of student performance expectation. SA learning activities might decrease initial students' performance but contribute to increased delayed performance (Archibald, 2010).

In addition, students should be provided with instructional support during the SA learning activities to maximize learning benefits of the SA tools. Researchers emphasize that successful implementation of SA in learning settings requires some engagement from the students, which can be facilitated by an instructor (Kawase et al., 2009; Razon et al., *submitted for publication*).

SA tools have been found beneficial for collaborative learning. Many researchers recommend to use small teams to not only highlight and annotate web-based reading materials but also to engage in more substantive learning activities. Specifically, teams of 2–3 people were found optimal for collaborative activities such as peer-critique, articles critique, and other evaluation activities (Mendenhall & Author, 2010).

In summary, empirical findings suggest that in addition to the mentioned learning and affective benefits that social annotation technology offers, there are many general advantages related to SA tools including re-finding tools, easy manipulation and organization of the annotations and resources, and sharing capabilities.

With regard to design and technical features of the social annotation tools, it is important to examine thoroughly the available social annotation software. First of all, "the annotation action must be effortless in all senses — easy to access and visualize, as few interactions as possible and in-context interactions" (Kawase et al., 2009, pp. 12–13). When the use of a technology becomes a burden, users abandon it in favor of other more convenient tools. Particularly an educational technology should be easy to use, so its add-on instructional value would not be outweighed by its usability challenges, which can cause extraneous cognitive load and distract students from the learning process.

Second, when choosing a particular SA tool, it is important to verify that the tool supports all desired electronic formats such as Word, Excel, PowerPoint, Adobe Acrobat documents, etc. When students need to work with multiple documents, they prefer a tool that supports all electronic formats. Working with unsupported document formats can complicate instructional assignments and result in negative attitudes toward the SA tool.

- To sum up, the following recommendations for the use of SA technology in educational settings have been gathered from the literature review: Provide adequate technology training both for teachers and for students before actual implementation of SA technology-supported instructional activities in class.
- Allow students to get used to work with the new technology (at least 4 h) before implementing important SA instructional activities in addition to training on how to use a SA tool.
- Provide students with instructional support during the SA learning activities.
- Use small teams (2–3 students) for collaborative SA activities.
- Choose an easy-to-use SA tool.
- Verify that a SA tool supports all desired electronic formats.

In conclusion, it is important to emphasize that like any other technology, SA tools have their advantages and disadvantages and may lead to learning gains if used in the right context, when needed, and while providing the required guidance.

6. Future research

Even though the literature review shows potential of social annotations tools in education, the use of this technology is still limited and has had minimal controlled evaluation trial and in-depth qualitative investigations in the education setting. Each of the following limitations is a new opportunity for research. First, more SA-related studies should be conducted in the educational setting. One of the major limitations in the existing research was small sample sizes. Therefore, to increase the quality of research findings larger sample sizes should be examined. Furthermore, it would be interesting if researchers were able to examine the changes in the learner's annotation behaviors over the time and how these changes affect learning outcomes, since

the process of learning progression (i.e., changing from novice in annotating to expert) plays an important role in improving learning and can affect learner's motivation and attitude toward learning as well.

In addition to the learning progression during the annotation learning process, it is important to explore the relationships between other students' traits (e.g., cognitive styles) and various SA instructional activities. Another interesting aspect would be integrating teamwork activities while using the social annotation tools and examining the effects of those tools on team mental models that are expected to enhance team-learning performance and in turn might affect individual learning.

To conclude, the integration of SA tools in education is evolving without sufficient evidence whether their use indeed enhances learning and motivation. Moreover, there is little understanding under which conditions and within which contexts these tools should be implemented. Therefore, conducting more research in this area is required in order to better understand the conditions, under which SA technology can lead to better learning outcomes and higher motivation.

References

- Archibald, T. N. (2010). *The effect of the integration of social annotation technology, first principles of instruction, and team-based learning on students' reading comprehension, critical thinking, and meta-cognitive skills*. PhD Dissertation, Florida State University, Tallahassee, FL.
- Bateman, S., Brooks, C., McCalla, G., & Brusilovsky, P. (2007). Applying collaborative tagging to e-learning. Paper presented at the Proceedings of the 16th International World Wide Web Conference (WWW2007).
- Bateman, S., Farzan, R., Brusilovsky, P., & McCalla, G. (2006, November 8–10). OATS: The open annotation and tagging system. Paper presented at the Third Annual International Scientific Conference of the Learning Object Repository Research Network, Montreal.
- Bottoni, P., Civica, R., Levialdi, S., Orso, L., Panizzi, E., & Trinchese, R. (2004). MADCOW: A multimedia digital annotation system. Paper presented at the Proceedings of the working conference on Advanced Visual Interfaces, Gallipoli, Italy.
- Bottoni, P., Levialdi, S., Labella, A., Panizzi, E., Trinchese, R., & Gigli, L. (2006, May 23–26). MADCOW: A visual interface for annotating web pages. Paper presented at the Working Conference on Advanced Visual Interfaces Venezia, Italy.
- Cooper, H. M. (1988). Organizing knowledge synthesis: A taxonomy of literature reviews. *Knowledge in Society*, 1, 104–126.
- Davis, J., & Huttenlocher, D. (1995). Shared annotation for cooperative learning. Paper presented at the CSCI'95.
- Elliot, A. J. (1997). Integrating the classic and contemporary approaches to achievement motivation: A hierarchical model of approach and avoidance achievement motivation. In M. Maehr, & P. Pintrich (Eds.), *Advances in motivation and achievement*, Vol. 10. (pp. 143–179) London: JAI Press.
- Elliot, A. J., & Murayama, K. (2008). On the measurement of achievement goals: Critique, illustration, and application. *Journal of Educational Psychology*, 100, 613–628.
- Gao, F., & Johnson, T. E. (in press). Learning Web-based materials collaboratively with a Web annotation tool.
- Glover, L., Xu, Z., & Hardaker, G. (2007). Online annotation – Research and practices. *Computers & Education*, 49, 1308–1320.
- Huang, Y.-M., Huang, T.-C., & Hsieh, M.-Y. (2008). Using annotation services in a ubiquitous jigsaw cooperative learning environment. *Educational Technology & Society*, 11(2), 3–15.
- Hwang, W.-Y., Wang, C.-Y., & Sharples, M. (2007). A study of multimedia annotation of web-based materials. *Computers & Education*, 48, 680–699.
- Jeong, H., & Hmelo-Silver, C. E. (2010). Productive use of learning resources in an online problem-based learning environment. *Computers in Human Behavior*, 26(1), 84–99.
- Johnson, T. E., Archibald, T. N., & Tenenbaum, G. (2010). Individual and team annotation effects on students' reading comprehension, critical thinking, and meta-cognitive skills. *Computers in Human Behavior*, 26(6), 1496–1507.
- Johnson, T. E., Khalil, M. K., & Spector, J. M. (2008). The role of acquired shared mental models in improving the process of team-based learning. *Educational Technology & Society*, 48(4), 18–26.
- Kawasaki, Y., Sasaki, H., Yamaguchi, H., & Yamaguchi, Y. (2008). Effectiveness of highlighting as prompt in text reading on computer monitor. Paper presented at the 8th WSEAS International Conference on Multimedia Systems and Signal Processing, Hangzhou: China.
- Kawase, R., Herder, E., & Nejdil, W. (2009). A comparison of paper-based and online annotations in the workplace. Paper presented at the Proceedings of the 4th European Conference on Technology Enhanced Learning: Learning in the Synergy of Multiple Disciplines, Nice, France.
- Kawase, R., Herder, E., & Nejdil, W. (2010, April 7–10). Annotations and hypertrails with Spreadcrumbs – An easy way to annotate, refine and share. Paper presented at the WEBIST – The 6th International Conference on Web Information Systems and Technologies, Valencia, Spain.
- Kirschner, F., Paas, F., & Kirschner, P. (2009). Individual and group-based learning from complex cognitive tasks: Effects on retention and transfer efficiency. *Computers in Human Behavior*, 25(2), 306–314.
- Laughlin, P. R., Bonner, B. L., & Miner, A. G. (2006). Groups perform better than the best individuals on letters-to-numbers problems: Effects of group size. *Journal of Personality and Social Psychology*, 90(4), 644–651.
- Lebow, D. G., & Lick, D. W. (2004). HyLighter: An effective interactive annotation innovation for distance education. Paper presented at the 20th Annual Conference on Distance Teaching and Learning.
- LeeTiernan, S., & Grudin, J. (2001). Fostering engagement in asynchronous learning through collaborative multimedia annotation. *Proceedings of INTERACT*, 2001, 472–479.
- Mendenhall, A., & Author, T. E. (2010). Title. *Interactive Learning Environments*.
- Mendenhall, A., & Johnson, T. E. (2010). Fostering the development of critical thinking skills, and reading comprehension of undergraduates using a Web 2.0 tool coupled with a learning system. *Interactive Learning Environments*, 18(3), 263–276.
- Merrill, M. D. (2002). First principles of instruction. *Educational Technology, Research and Development*, 50(2), 43–59.
- Nokelainen, P., Kurhila, J., Miettinen, M., Floreen, P., & Tirri, H. (2003). Evaluating the role of a shared document-based annotation tool in learner-centered collaborative learning. Paper presented at the Advanced Learning Technologies. The 3rd IEEE International Conference.
- Nokelainen, P., Miettinen, M., Kurhila, J., Floréen, P., & Tirri, H. (2005). A shared document-based annotation tool to support learner-centred collaborative learning. *British Journal of Educational Technology*, 36(5), 757–770. doi:10.1111/j.1467-8535.2005.00474.x.
- Pintrich, P. R., Smith, D. A., Garcia, T., & McKeachie, W. J. (1991). *A manual for the use of the motivated strategies for learning questionnaire (MSLQ)*. Ann Arbor: National Center for Research to Improve Postsecondary teaching and Learning, University of Michigan.
- Pintrich, P. R., Smith, D. A., Garcia, T., & McKeachie, W. J. (1993). Reliability and predictive validity of the motivated strategies for learning questionnaire (MSLQ). *Educational and Psychological Measurement*, 53, 801–813.
- Razon, S., Turner, J., Johnson, T. E., Arsal, G., & Tenenbaum, G. (submitted for publication). Effects of a collaborative annotation method on students' learning and learning-related motivation and affect. *Computers in Human Behavior*.
- Samuel, R., Kim, C., & Johnson, T. E. (in press). A study of online learning within a social annotation modeling learning system. *Journal of Educational Computing Research*.
- Wu-Yuin, H., Chin-Yu, W., & Mike, S. (2007). A study of multimedia annotation of Web-based materials. *Computers & Education*, 48(4), 680–699. doi:10.1016/j.compedu.2005.04.020.
- Yang, S. J. H., Chen, I. Y. L., & Shao, N. W. Y. (2004). Ontology enabled annotation and knowledge management for collaborative learning in virtual learning community. *Educational Technology & Society*, 7(4), 70–81.
- Zellweger, P., Mangen, A., & Newman, P. (2002). Authoring fluid narrative hypertexts using treatable visualizations. Paper presented at the ACM Hypertext.

Elena Novak is a research assistant and instructor at Florida State University. Elena Novak is also a PhD candidate in Instructional Systems, at the Department of Educational Psychology and Learning Systems, Florida State University. Her research interests include instructional games and simulations, gaming characteristics, storyline, emotion assessment, active learning, and human performance improvement.

Rim Razzouk completed her PhD in the Department of Educational Psychology and Learning Systems at the Florida State University, Tallahassee, FL. Her research focuses on effects of case studies on students' learning outcomes, attitudes, and team-shared mental models in a team-based learning environment. Her research interests include technology integration in education; assessment and evaluation; learner-centered methods and strategies; and any other methods that assist in enhancing human performance.

Tristan E. Johnson is on the faculty with both the Learning Systems Institute and the Instructional Systems program in the Department of Educational Psychology and Learning Systems at Florida State University. Dr. Johnson has several years of experience studying team cognition, team-based learning, measuring shared mental models, team assessment and diagnostics, and team interventions. He is also involved in the development of agent-based modeling and the creation of data-driven algorithms for modeling training and instructional effects within performance models.